Effects of Stitching, Layup Orientation, and Core Backing on Kevlar Ballistic Performance

Anna Wilson, Hannah Redoblado, Jamie Ebbs, and Eltahry Elghandour

Department of Mechanical Engineering, California Polytechnic State University
1 Grand Ave, San Luis Obispo, CA 93407

Ballistic protection using woven aramid fabrics depends not only on the number and orientation of fabric layers, but also on how the layers are joined and supported. This study investigated whether sewing patterns, stacking sequence, and core backing influence the ballistic performance and deflection of layered Kevlar panels. Woven aramid Kevlar was arranged using 0°/90° and ±45° orientations. Twenty-layer Kevlar laminate stacks were produced by alternating two layers of 0°/90° fabric with two layers of ±45° fabric, creating a symmetric layup. Two stitching configurations were evaluated: stitching aligned only at 0°/90°, and stitching incorporating both 0°/90° and ±45° directions. The laminated raw-material samples measured 5.5 × 5.5 in. and were sewn with V-138 thread at 4.5 mm stitch spacing, which has a manufacturer-rated breaking strength of 21.5 lbs. Panels were suspended from a paper-mâché mannequin and fired upon with 9 mm rounds from 5 m. The laminated raw-material samples with sewn configurations exhibited substantially less backface deformation while stopping the 9 mm projectile. Unbacked 0°/90° samples exhibited minimal protrusion and survived up to three rounds before significant deformation, while unbacked ±45° samples survived four rounds. Samples incorporating core backing distributed impact forces more evenly, survived multiple rounds with minimal backlash, and produced only minimally invasive core damage, with core bending measured up to 0.5 mm. These findings indicate that stitch orientation, layer orientation, and core support can materially influence the multi-hit response and deformation behavior of sewn Kevlar systems. The results support further optimization of sewn aramid layups for lightweight ballistic protection where reduced deformation and improved force distribution are desired.